

Mini-case 2.1

Variance:

① I. $\bar{r}_A = 0,25 \cdot 0,2 + 0,65 \cdot 0,05 + 0,1 \cdot (-0,3125) = 0,05125$

II. $\bar{r}_B = 0,7 \cdot 0,1375 + 0,3 \cdot (-0,15) = 0,05125$

" $\bar{r}$ " is mean value of return from an investment into a company.

② I. $\text{Var}(\tilde{r}_A) = \sigma_{\tilde{r}_A}^2 = E[(\tilde{r}_A - \bar{r}_A)^2] = 0,25(0,2 - 0,05125)^2 + 0,65(0,05 - 0,05125)^2 + 0,1(-0,1 - 0,05125)^2 = 0,00782013 \approx 0,0078$

II. $\text{Var}(\tilde{r}_B) = \sigma_{\tilde{r}_B}^2 = E[(\tilde{r}_B - \bar{r}_B)^2] = 0,7(0,1375 - 0,05125)^2 + 0,3(-0,15 - 0,05125)^2 = 0,017357 \approx 0,017$

Standard deviation:

① I. $\sigma_{\tilde{r}_A} = \sqrt{\sigma_{\tilde{r}_A}^2} = 0,08843 \approx 0,09$

II. $\sigma_{\tilde{r}_B} = \sqrt{\sigma_{\tilde{r}_B}^2} = 0,13174 \approx 0,13$

Mini-case 2.2.

$$FV(\text{date 1}) = \begin{cases} 15000 \text{ €} & , p_1 = 0,5 \\ 7000 \text{ €} & , p_2 = 0,5 \end{cases}$$

$$U(w) = \frac{w^{1-\gamma} - 1}{1-\gamma}, \text{ whereby } \gamma \text{ is a coefficient of relative risk aversion.}$$

$$\gamma_A = 0,75 ; \gamma_B = 0,5$$

a) I. $E(w) = 0,5 \cdot 15000 + 7000 \cdot 0,5 = 11000 \text{ €}$ for everyone of both investors

II. $E[U(w)] :$

$$1) U_A(w) = \frac{w^{1-\gamma} - 1}{1-\gamma} = \frac{w^{0,25} - 1}{0,25}$$

$$\vdots \gamma_A = 0,75$$

$$E_A[U(w)] = 0,5 \frac{15000^{0,25} - 1}{0,25} + 0,5 \frac{7000^{0,25} - 1}{0,25} = 36,427$$

$$2) U_B(w) = \frac{w^{1-\gamma} - 1}{1-\gamma} = \frac{w^{0,5} - 1}{0,5}$$

$$\vdots \gamma_B = 0,5$$

$$E_B[U(w)] = 0,5 \frac{15000^{0,5} - 1}{0,5} + 0,5 \frac{7000^{0,5} - 1}{0,5} = 204,14$$

b) The lowest amount the investors are willing to sell their positions for right now instead of holding them until the next period is preimage of expected utility of wealth also known as certainty equivalent

$$1. U_A(w) = \frac{w^{0,25} - 1}{0,25} \quad | \cdot 0,25$$

$$0,25 U_A(w) = w^{0,25} - 1 \quad | +1$$

$$0,25 U_A(w) + 1 = w^{0,25} \quad | \sqrt[0,25]{}$$

$$W_A = CE_A = \sqrt[0.25]{0.25 U_A(W) + 1} = \underline{10433.893}$$

$$II. U_B = \frac{W^{0.5} - 1}{0.5}$$

$$W = \sqrt[0.5]{0.5 \cdot U_B(W) + 1} = \underline{10623.42}$$

Ⓒ According to the computations above, only first investor would sell his position under such circumstances.

Mini-case 2.3.

Ⓐ I. $\bar{r}_A = (0.07 + 0.1 + 0.005 + 0.06) : 4 = \underline{0.05875}$

II. $\bar{r}_B = (0.1 + 0.04 + (-0.05) + 0.01) : 4 = \underline{0.025}$

III. $\text{cov}(\tilde{r}_A, \tilde{r}_B) = \frac{1}{4} [(0.005 - \bar{r}_A)(-0.05 - \bar{r}_B)] + \frac{1}{4} [(0.1 - \bar{r}_A) \cdot (0.04 - \bar{r}_B)] + \frac{1}{4} [(0.005 - \bar{r}_A)(-0.05 - \bar{r}_B)] + \frac{1}{4} [(0.06 - \bar{r}_A) \cdot (0.01 - \bar{r}_B)] = \underline{0.00136875}$

Ⓑ To determine degree of co-movement we compute correlation for the two securities

$$\rho(\tilde{r}_A, \tilde{r}_B) = \frac{\text{cov}(\tilde{r}_A, \tilde{r}_B)}{\sigma_A \sigma_B} = \underline{0.73684}$$

$$\sigma_A^2 = \frac{1}{4} [(0.07 - \bar{r}_A)^2 + (0.1 - \bar{r}_A)^2 + (0.005 - \bar{r}_A)^2 + (0.06 - \bar{r}_A)^2] = \underline{0.0011796}$$

$$\sigma_B^2 = \frac{1}{4} [(0.1 - \bar{r}_B)^2 + (0.04 - \bar{r}_B)^2 + (-0.05 - \bar{r}_B)^2 + (0.01 - \bar{r}_B)^2] = \underline{0.002925}$$

$$\sigma_A = 0.0343; \quad \sigma_B = 0.054083$$

Strong positive correlation. Returns move together. The 'stocks' diversification effects are restricted to quite small

Mini-case 2.6

$$\textcircled{a} \quad \mathbb{E}\{U(\tilde{R}_p)\} = \mathbb{E}(\tilde{R}_p) - \frac{\gamma}{2} \mathbb{E}[(\tilde{R}_p - \mathbb{E}(\tilde{R}_p))^2] = \mathbb{E}(\tilde{R}_p) - \frac{\gamma}{2} \text{var}(\tilde{R}_p)$$

$$U(\tilde{R}_p) = \tilde{R}_p - \frac{\gamma}{2} (\tilde{R}_p - \mathbb{E}(\tilde{R}_p))^2 = \tilde{R}_p - \frac{\gamma}{2} \text{var}(\tilde{R}_p)$$

1) $\tilde{R} = \sum_{i=1}^N x_i \tilde{r}_i = x_1 r_f + x_2 \tilde{r}_s$, where r_f is return of a risk-free bond and $\tilde{r}_s$ is for the risky stock

2) $U(\tilde{R}_p = x_1 r_f + x_2 \tilde{r}_s) = x_1 r_f + x_2 \tilde{r}_s - \frac{\gamma}{2} [(x_1 r_f + x_2 \tilde{r}_s) - \mathbb{E}(x_1 r_f + x_2 \tilde{r}_s)]^2 = (x_1 r_f + x_2 \tilde{r}_s) - \frac{\gamma}{2} \text{var}(x_1 r_f + x_2 \tilde{r}_s)$

$$x_1 = 1 - x_2$$

$$\mathbb{E}(\tilde{R}_p) = x_1 \mathbb{E}(r_f) + x_2 \mathbb{E}(\tilde{r}_s) = x_1 r_f + x_2 \mathbb{E}(\tilde{r}_s)$$

III. $U(\tilde{R}_p = (1-x_2)r_f + x_2 \tilde{r}_s) = (1-x_2)r_f + x_2 \tilde{r}_s - \frac{\gamma}{2} [(1-x_2)r_f + x_2 \tilde{r}_s - \mathbb{E}[(1-x_2)r_f + x_2 \tilde{r}_s]]^2 = [(1-x_2)r_f + x_2 \tilde{r}_s] - \frac{\gamma}{2} \text{var}[(1-x_2)r_f + x_2 \tilde{r}_s]$

IV. $U(\tilde{R}_p = r_f - x_2(r_f - \tilde{r}_s) = r_f - x_2(r_f - \tilde{r}_s) - \frac{\gamma}{2} x_2^2 \text{var}(\tilde{r}_s)$

V. $U(\tilde{R}_p = r_f - r_f x_2 - x_2 \tilde{r}_s) = r_f - x_2(r_f - \tilde{r}_s) - \frac{\gamma}{2} x_2^2 \text{var}(\tilde{r}_s)$

VI. $\frac{dU}{dx_2} = -1(r_f - \tilde{r}_s) - 2x_2 \frac{\gamma}{2} \text{var}(\tilde{r}_s) = -r_f + \tilde{r}_s - x_2 \gamma \cdot \text{var}(\tilde{r}_s)$

$$-r_f + \tilde{r}_s - x_2 \gamma \cdot \text{var}(\tilde{r}_s) = 0$$

$$-x_2 \cdot \gamma \cdot \text{var}(\tilde{r}_s) = r_f - \tilde{r}_s$$

$$-x_2 = \frac{r_f - \tilde{r}_s}{\gamma \text{var}(\tilde{r}_s)}$$

$$x_2 = -\frac{r_f - \tilde{r}_s}{\gamma \text{var}(\tilde{r}_s)} = \frac{\tilde{r}_s - r_f}{\gamma \text{var}(\tilde{r}_s)}$$

⑥ When increasing, γ coefficient of relative risk aversion takes the portfolio weight x_2 down. The lower γ value is, the smaller divisor of $\bar{r}_s - r_f$ is going to be, so that x_2 gets bigger. This makes much of sense, as it means that portfolio weight of the risky investment gets bigger with a lower risk aversity.